

untangle::async

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Chapter 1

async

c++ mechanism to run queued callables in an asynchronous fashion

The interface consists of one header file that exposes a generic execution class. It is a wrapper of a list of callables that are invoked on other thread. The callables are pumped into the list by invoking an action from the caller thread.

The implementation is using the **actuator** callable, and therefore it exemplifies how a generic callable may improve the code structure, by easily creating interfaces inside a class instead of using external ones.

There are provided two execution modes: one-off and continuous.

requirements

C++23. The interface reports warnings with `std::println`, so it needs a standard library that provides `<print>` — GCC 14 or newer; older `libc++` and `MSVC` releases may not have it yet.

one-off execution



example: one-off execution of a plain function

```
#include <async.hpp>

void f()
{
    std::cout << "f thread " << std::this_thread::get_id() << std::endl;
}

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    // declare an execution specialized for an action type; it must be held by a shared pointer,
    // because an async binding keeps only weak ownership of it
    // the name is optional, and identifies the execution in the warnings it reports
    using void_execution = untangle::async::execution<std::function<void(void)>>;
    auto execution = void_execution::create_instance("oneoff");
```

```

std::function<void(void)> action;
// create async binding between the action and f; the execution is passed as a shared pointer,
// which the binding holds weakly - that is what lets the action outlive it safely
void_execution::bind_action_and_function(action, f, execution);
// whenever the action is invoked, a new callable wrapping f will be added to the execution's action list
action();
// run the execution
execution->run();

// add the execution object to the running poll
untangle::async::execution_poll::get().add(*execution);

// wait the polled executions to finish
while(untangle::async::execution_poll::get().is_running())
{
    std::this_thread::sleep_for(std::chrono::milliseconds(1));
}

return 0;
}

```

example: one-off execution of a class method

```

#include <async.hpp>

class A
{
public:
    void f(int x)
    {
        std::cout << "A::f " << x << " thread " << std::this_thread::get_id() << std::endl;
    }
    std::function<void(int)> action;
};

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    auto a = std::make_shared<A>();

    // declare an execution specialized for an action type; it must be held by a shared pointer
    using int_execution = untangle::async::execution<std::function<void(int)>>;
    auto execution = int_execution::create_instance("oneoff_method");
    // note both the bound object and the execution must be shared pointers
    int_execution::bind_action_and_method(a->action, a, &A::f, execution);
    // whenever the action is invoked, a new callable wrapping f will be added to the execution's action list
    a->action(10);
    // run the execution
    execution->run();

    // add the execution object to the running poll
    untangle::async::execution_poll::get().add(*execution);

    // wait the polled executions to finish
    while(untangle::async::execution_poll::get().is_running())
    {
        std::this_thread::sleep_for(std::chrono::milliseconds(1));
    }

    return 0;
}

```

continuous execution



example: continuous execution of a plain function

```

#include <async.hpp>

void f()
{
    std::cout << "f thread " << std::this_thread::get_id() << std::endl;
}

```

```

}

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    // declare an execution specialized for an action type; it must be held by a shared pointer
    using void_execution = untangle::async::execution<std::function<void(void)>>;
    auto execution = void_execution::create_instance("continuous");
    std::function<void(void)> action;
    // create async binding between the action and f
    void_execution::bind_action_and_function(action, f, execution);

    // run the execution
    execution->start();
    // add f to the execution's queue (three times)
    action();action();action();
    // stop the execution
    execution->stop();

    // add the execution object to the running poll
    untangle::async::execution_poll::get().add(*execution);

    // wait the polled executions to finish
    while(untangle::async::execution_poll::get().is_running())
    {
        std::this_thread::sleep_for(std::chrono::milliseconds(1));
    }

    return 0;
}

```

queueing and lifetime

Queueing is thread safe. An action may be invoked from any thread; the execution's list is guarded, and the worker is woken rather than polled for.

`stop()` ends the execution's working life. The worker runs whatever was queued before the call and then leaves. An action invoked after `stop()` is refused rather than queued, because it would never run, and the refusal is reported:

```
warning: execution 'continuous' is stopped, action not added
```

An action bound to an object that has since been destroyed is dropped the same way, rather than ending the process:

```
warning: execution 'continuous' dropped an invalid action: bind: invalid object
```

The worker is detached and cannot be joined. Waiting for an execution therefore means waiting on its running state, which is what `execution_poll` reports and what the examples above do. The destructor waits on the same state, so an execution that goes out of scope while its worker is still running blocks until the worker is finished rather than leaving it reading freed memory.

The poll holds a pointer to each execution added to it, so an execution withdraws itself in its destructor. `execution_poll::remove()` is available for withdrawing one earlier; calling it is not required.

building the tests

```

cmake -S test -B test/build
cmake --build test/build
ctest --test-dir test/build

```

`test/CMakeLists.txt` fetches googletest at configure time, so the first configure needs network access. The actuator submodule has to be present: `git submodule update --init`.

sanitizers

ASYNC_SANITIZE builds the tests under a sanitizer. It is off by default, because a sanitized build is several times slower and ThreadSanitizer does not ship for every toolchain.

```
cmake -S test -B test/build-asan -DASYNC_SANITIZE=address
cmake --build test/build-asan
ctest --test-dir test/build-asan
```

Accepted values are address, thread, undefined, or empty. Anything else is refused at configure time rather than passed through to the compiler.

Use a separate build directory per sanitizer. address and thread instrument the same accesses in incompatible ways and cannot be combined, so one build directory is one sanitizer.

The two answer different questions. ThreadSanitizer names both sides of a data race, which is how the queue's races were diagnosed. AddressSanitizer reports a use-after-free, which is how the dangling attachment was. Neither sees everything: a defect reached through a destroyed `std::mutex` throws `std::system_error` from inside `libsystem` before any instrumented load runs, so both sanitizers stay silent. Reach for a direct instrumented read when that happens.

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

namespace untangle: functions [13](#)

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle	Interface to untangle::async::execution class	15
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Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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untangle::async::execution< actionT >	
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Chapter 5

File Index

5.1 File List

Here is a list of all documented files with brief descriptions:

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Chapter 6

Topic Documentation

6.1 namespace untangle: functions

Chapter 7

Namespace Documentation

7.1 untangle Namespace Reference

Interface to [untangle::async::execution](#) class.

7.1.1 Detailed Description

Interface to [untangle::async::execution](#) class.

Chapter 8

Class Documentation

8.1 `untangle::async::attachment` Struct Reference

One execution's place in the attachment graph: who attached it, if anyone.

```
#include <async.hpp>
```

Public Attributes

- `std::weak_ptr< attachment >` `attacher`
The link of the execution that attached this one; empty when nothing has.

8.1.1 Detailed Description

One execution's place in the attachment graph: who attached it, if anyone.

At namespace scope, so it is one type rather than one per specialisation: an execution may be attached by an execution of any specialisation, and the chain is walked without knowing any link's action type. It dies with its execution, so it doubles as a liveness token - a link that still locks belongs to an execution that is still there.

The documentation for this struct was generated from the following file:

- `async.hpp`

8.2 `untangle::async::execution< actionT >` Class Template Reference

Asynchronous template execution class.

```
#include <async.hpp>
```

Public Member Functions

- `execution ()`
Constructs an execution with the default name. Call `create_instance()` instead.
- `execution (std::string exec_name)`
Constructs a named execution. Use `create_instance()` for one that is to be bound.
- `~execution ()`
Destroys the execution, once its worker has left.
- `bool is_running () const`
Is a worker of this execution's still there?
- `bool is_busy () const`
Is this execution working through its action list?
- `void run ()`
Runs what is queued on a worker of its own, once, and keeps the results.
- `void start ()`
Starts a continuous worker that runs whatever is queued until `stop()`.
- `void stop ()`
Stops the worker, after the actions already queued have run.
- `template<typename... Args>`
`bool add_action (actionT action, Args &&... args)`
Queues an action, bound to args, and says whether it was taken.
- `template<typename otherT>`
`void attach (const std::shared_ptr< otherT > &other)`
Attaches another execution, so that this one triggers it.
- `template<typename otherT>`
`bool detach (otherT &other)`
Detaches an execution previously attached by `attach()`.
- `auto results () const`
The return values of the actions run by `run()`, in the order they ran.

Static Public Member Functions

- `template<typename... Args>`
`static std::shared_ptr< execution > create_instance (Args &&... args)`
Creates an execution owned by a `std::shared_ptr`, which is what binding to it requires.
- `template<typename classT, typename T>`
`static void bind_action_and_method (actionT &action, const std::shared_ptr< classT > &obj, T classT::*method, const std::shared_ptr< execution > &async_exec)`
Binds asynchronously an external action to a class function member.
- `template<typename T>`
`static void bind_action_and_function (actionT &action, T Fn, const std::shared_ptr< execution > &async_exec)`
Binds asynchronously an external action to a plain function.

Public Attributes

- `std::function< void(void)> action_execute`
Connection point that runs this execution's queued actions. Bound to `execute_actions()`.
- `std::function< void(void)> action_stop`
Connection point that stops this execution. Bound to `stop()`.
- `std::function< bool(void)> action_is_running`
Connection point that reports whether this execution is running. Bound to `is_running()`.
- `std::function< void(void)> on_finished`
Called on the worker's thread once the action list has drained. Assigned by the caller.

Static Public Attributes

- `static constexpr auto default_name = "default_name"`
Name of an execution built unnamed.

8.2.1 Detailed Description

```
template<typename actionT>
class untangle::async::execution< actionT >
```

Asynchronous template execution class.

An async execution provides a mechanism to queue actions and execute them sequentially on a separate thread. It may also attach another [execution](#) object and trigger its actions. This way actions of different types may be executed on the same thread.

The mechanism relies on an "asynchronous binding" created by [bind_action_and_method\(\)](#) or [bind_action_and_function\(\)](#).

Template Parameters

<code>actionT</code>	It represents the type of the action. It is specified as <code>std::function<...></code> and should match the signature of the bound function or class method.
----------------------	--

8.2.2 Constructor & Destructor Documentation

8.2.2.1 `execution()` [1/2]

```
template<typename actionT>
untangle::async::execution< actionT >::execution () [inline]
```

Constructs an execution with the default name. Call [create_instance\(\)](#) instead.

The name identifies this execution in the warnings it reports, and is otherwise unused.

8.2.2.2 `execution()` [2/2]

```
template<typename actionT>
untangle::async::execution< actionT >::execution (
    std::string exec_name) [inline], [explicit]
```

Constructs a named execution. Use [create_instance\(\)](#) for one that is to be bound.

Parameters

exec_name	- A name for this execution.
-----------	------------------------------

8.2.2.3 ~execution()

```
template<typename actionT>
untangle::async::execution< actionT >::~~execution () [inline]
```

Destroys the execution, once its worker has left.

The worker is detached and cannot be joined, so this waits on the "running" state instead. It detaches this execution from whatever attached it first, and returns at once for one that was never started.

Attention

Safe only because running_ is cleared as the very last thing the worker does: this object may be freed the moment it reads false, so nothing may follow it in execute() or loop().

Remarks

No rule of five, deliberately. An execution is declared once and used in place; the members already delete the copy operations several times over.

8.2.3 Member Function Documentation

8.2.3.1 create_instance()

```
template<typename actionT>
template<typename... Args>
std::shared_ptr< execution > untangle::async::execution< actionT >::create_instance (
    Args &&... args) [inline], [static]
```

Creates an execution owned by a std::shared_ptr, which is what binding to it requires.

An execution built any other way is still usable - `add_action()` takes actions directly - but cannot be bound, and the attempt does not compile.

Parameters

args	- Constructor arguments: a name, or nothing for the default name.
------	---

Returns

- A std::shared_ptr owning the new execution.

8.2.3.2 bind_action_and_method()

```
template<typename actionT>
template<typename classT, typename T>
void untangle::async::execution< actionT >::bind_action_and_method (
    actionT & action,
    const std::shared_ptr< classT > & obj,
    T classT::* method,
    const std::shared_ptr< execution< actionT > > & async_exec) [inline], [static]
```

Binds asynchronously an external action to a class function member.

It creates an action as a binding to a class method (by untangle::bind()), and assigns to action a callable that queues it. The execution is held weakly, so the action may outlive it; the action itself is shared with every call queued from it, rather than copied per call.

Attention

Invoking action after the execution has been destroyed throws untangle::invalid_action. The worker catches it and drops the action with a warning.

Parameters

action	[in,out] - An action of type std::function<...>.
obj	- A std::shared_ptr that wraps the bound class object.
method	- Pointer to function member. It is specified as &<class type>::<function member>.
async_exec	- A std::shared_ptr owning the execution the action is queued on. Requiring it here is what keeps an execution that no shared_ptr owns from being bound at all.

8.2.3.3 bind_action_and_function()

```
template<typename actionT>
template<typename T>
void untangle::async::execution< actionT >::bind_action_and_function (
    actionT & action,
    T Fn,
    const std::shared_ptr< execution< actionT > > & async_exec) [inline], [static]
```

Binds asynchronously an external action to a plain function.

It wraps Fn in an action and assigns to action a callable that queues it. The execution is held weakly, as in [bind_action_and_method\(\)](#), and the action is shared with every call queued from it rather than copied per call.

Parameters

action	[in,out] - An action of type std::function<...>.
Fn	- A plain function, or any callable; moved into the binding.

async_exec	- A std::shared_ptr owning the execution the action is queued on.
------------	---

8.2.3.4 is_running()

```
template<typename actionT>
bool untangle::async::execution< actionT >::is_running () const [inline]
```

Is a worker of this execution's still there?

True from `run()` or `start()` until the worker has left, which is the same instant `~execution()` waits for and the same one `run()` waits for before starting another.

Remarks

`on_finished` is raised from inside the drain and reads this as true - the worker is standing in the callback with the rest of its path still to go.

Returns

true - a worker is still there.

8.2.3.5 is_busy()

```
template<typename actionT>
bool untangle::async::execution< actionT >::is_busy () const [inline]
```

Is this execution working through its action list?

True from the moment an action is added until the last one has returned. Unlike `is_running()`, which is about the worker thread, this answers "has it room for more work"

- a worker started by `start()` is running for its whole life whether busy or idle.

Remarks

Answers for this execution's own actions, not for those it triggers through `attach()`.

A false answer is durable only for the caller that is itself adding the actions; a true one may stop being true the moment after.

Returns

true - actions are queued, or one is running.

8.2.3.6 run()

```
template<typename actionT>
void untangle::async::execution< actionT >::run () [inline]
```

Runs what is queued on a worker of its own, once, and keeps the results.

The worker is detached: wait for it with `is_running()` or `execution_poll`. Actions queued while it is draining are run by it too. `results()` is filled by this path only.

Blocks until a worker still resident from a previous run has left, so one execution never has two. `is_running()` reads false at exactly that moment, so a caller that waits for it and then runs again does not wait here at all.

Attention

Never call this from inside an action: the worker that would have to leave is the one making the call, so it waits for itself and hangs.

8.2.3.7 start()

```
template<typename actionT>
void untangle::async::execution< actionT >::start () [inline]
```

Starts a continuous worker that runs whatever is queued until `stop()`.

Unlike `run()`, the worker stays for the life of the execution, so `is_running()` is true whether or not there is work; ask `is_busy()` instead. Results are not collected.

Blocks until a worker still resident has left, as `run()` does.

Attention

Never call this from inside an action, for the reason given on `run()`.

8.2.3.8 stop()

```
template<typename actionT>
void untangle::async::execution< actionT >::stop () [inline]
```

Stops the worker, after the actions already queued have run.

Remarks

An action passed to `add_action()` after this call is refused, not queued: `stop()` is the end of this execution's working life, and an action accepted then would never run.

8.2.3.9 add_action()

```
template<typename actionT>
template<typename... Args>
bool untangle::async::execution< actionT >::add_action (
    actionT action,
    Args &&... args) [inline]
```

Queues an action, bound to args, and says whether it was taken.

It runs later, on whichever thread drives this execution - its own worker, or an attacher's. Queueing before [run\(\)](#) or [start\(\)](#) is normal.

Attention

A stopped execution refuses and drops the action, silently for a caller that ignores the answer. Nothing is thrown: `untangle::invalid_action` means a dead binding, while a refused action is sound and merely too late.

Remarks

A caller arriving through [bind_action_and_method\(\)](#) or [bind_action_and_function\(\)](#) never sees the answer: those lambdas return `actionT::result_type`, which has no room for it.

Template Parameters

Args	- The argument types the action is bound to.
------	--

Parameters

action	- The action to queue.
args	- The arguments to bind to action.

Returns

true - queued, and this execution's worker will run it.
false - the execution is stopped and the action was dropped.

8.2.3.10 attach()

```
template<typename actionT>
template<typename otherT>
void untangle::async::execution< actionT >::attach (
    const std::shared_ptr< otherT > & other) [inline]
```

Attaches another execution, so that this one triggers it.

The attached execution's pending actions are run by this one's worker, and stopping this one stops that one too. Actions of different types can be run on one thread this way.

Parameters

other	- A std::shared_ptr owning the execution to trigger; requiring one keeps an unowned execution from being attached at all. Attaching the same execution twice adds it twice, and one detach() still removes it completely.
-------	---

Exceptions

invalid_attachment	- if other is attached already. An execution has at most one attacher, which keeps the graph a forest and the cycle check a walk rather than a search.
invalid_attachment	- if the attachment would close a cycle, itself included. Driving a cycle recurses until the stack is gone, so it is refused where the caller can still be told.

8.2.3.11 detach()

```
template<typename actionT>
template<typename otherT>
bool untangle::async::execution< actionT >::detach (
    otherT & other) [inline]
```

Detaches an execution previously attached by attach().

Both paths are unwired together, the triggering one and the stopping one.

Parameters

other	- The execution to stop triggering. Detaching one that was never attached is not an error: the caller asked for a state that already holds.
-------	---

Returns

true - other was attached and is no longer.
false - other was not attached to this execution.

8.2.3.12 results()

```
template<typename actionT>
auto untangle::async::execution< actionT >::results () const [inline]
```

The return values of the actions run by run(), in the order they ran.

Read it from on_finished, which fires once the queue has drained, so the results are complete by the time the callback sees them. Returned by value: the worker is appending to it.

Remarks

Only run() fills this. A continuous worker has no point at which a run is over, so start() collects nothing.

Returns

The results of the most recent run, or empty if none has produced any.

8.2.4 Member Data Documentation

8.2.4.1 `action_execute`

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::action_execute
```

Connection point that runs this execution's queued actions. Bound to `execute_actions()`.

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.2 `action_stop`

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::action_stop
```

Connection point that stops this execution. Bound to `stop()`.

`attach()` wires it together with `action_execute`, and `detach()` unwires both.

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.3 `action_is_running`

```
template<typename actionT>
std::function<bool(void)> untangle::async::execution< actionT >::action_is_running
```

Connection point that reports whether this execution is running. Bound to `is_running()`.

`execution_poll` holds its address, and `~execution()` takes it back out.

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.4 `on_finished`

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::on_finished
```

Called on the worker's thread once the action list has drained. Assigned by the caller.

`results()` is complete by the time it runs, and `is_running()` reads true there.

The documentation for this class was generated from the following file:

- `async.hpp`

8.3 untangle::async::execution_poll Class Reference

Single-tone poll that reports whether any [execution](#) added to it is still running.

```
#include <async.hpp>
```

Public Member Functions

- `template<typename asyncexecT>`
`void add (asyncexecT &async_exec)`
 Adds an [execution](#) object to the poll.
- `template<typename asyncexecT>`
`void remove (asyncexecT &async_exec)`
 Removes an [execution](#) object from the poll.
- `auto is_running ()`
 Checks if the execution objects are running.

Static Public Member Functions

- `static execution_poll & get ()`
 Gets the single-tone instance.

8.3.1 Detailed Description

Single-tone poll that reports whether any [execution](#) added to it is still running.

Remarks

An execution runs in a detached thread and cannot be joined, so checking the "running" state is the only way to wait for one. This checks any number of them at once.

8.3.2 Member Function Documentation

8.3.2.1 add()

```
template<typename asyncexecT>
void untangle::async::execution_poll::add (
    asyncexecT & async_exec) [inline]
```

Adds an [execution](#) object to the poll.

Parameters

<code>async_exec</code>	An execution object.
-------------------------	--------------------------------------

Parameters

<code>async_exec</code>	An execution object.
-------------------------	--------------------------------------

8.3.2.3 `is_running()`

```
auto untangle::async::execution_poll::is_running () [inline]
```

Checks if the execution objects are running.

Returns

true - if at least one [execution](#) object in this poll is running.

8.3.2.4 `get()`

```
execution_poll & untangle::async::execution_poll::get () [inline], [static]
```

Gets the single-tone instance.

Returns

[execution_poll](#) An [execution_poll](#) instance.

The documentation for this class was generated from the following file:

- `async.hpp`

8.4 `untangle::async::invalid_attachment` Struct Reference

Raised by [execution::attach\(\)](#) when the attachment asked for is refused.

```
#include <async.hpp>
```

Public Member Functions

- [invalid_attachment](#) (std::string text)
Construct a new invalid attachment object.
- const char * [what](#) () const noexcept override
The message text describing the reason of this exception.

8.4.1 Detailed Description

Raised by [execution::attach\(\)](#) when the attachment asked for is refused.

Generated by Doxygen

Remarks

Not an `untangle::invalid_action`: that one reports a binding whose target has died and is swallowed by the kernel, while this one reports a binding whose target has died and is not swallowed

Parameters

text	- A message text, describing the reason of this exception.
------	--

8.4.3 Member Function Documentation

8.4.3.1 what()

```
const char * untangle::async::invalid_attachment::what () const    [inline], [override], [noexcept]
```

The message text describing the reason of this exception.

Returns

const char* - The message text.

The documentation for this struct was generated from the following file:

- async.hpp

Chapter 9

File Documentation

9.1 async.hpp

```
00001 // Copyright (c) 2018 Nicolae Popescu. MIT License.
00002
00006 #pragma once
00007
00008 #include <actuator/actuator.hpp>
00009 #include <algorithm>
00010 #include <atomic>
00011 #include <chrono>
00012 #include <condition_variable>
00013 #include <exception>
00014 #include <functional>
00015 #include <list>
00016 #include <memory>
00017 #include <mutex>
00018 #include <print>
00019 #include <string>
00020 #include <thread>
00021 #include <type_traits>
00022 #include <utility>
00023 #include <vector>
00024
00025 namespace untangle {
00026 namespace async {
00030
00037 struct invalid_attachment : std::exception {
00043     explicit invalid_attachment(std::string text) : message_(std::move(text)) {}
00044
00050     const char* what() const noexcept override { return message_.c_str(); }
00051
00052 private:
00053     std::string message_;
00054 };
00055
00064 struct attachment {
00066     std::weak_ptr<attachment> attacher;
00067 };
00068
00075 class execution_poll {
00076 public:
00082     template <typename asyncexecT>
00083     void add(asyncexecT& async_exec) {
00084         std::lock_guard<std::mutex> lock(actuator_mutex_);
00085
00086         if (!actuator_is_running_.is_connected()) {
00087             actuator_is_running_ = untangle::connect(async_exec.action_is_running);
00088         } else {
00089             actuator_is_running_.add(&async_exec.action_is_running);
00090         }
00091     }
00092
00101     template <typename asyncexecT>
00102     void remove(asyncexecT& async_exec) {
00103         std::lock_guard<std::mutex> lock(actuator_mutex_);
00104
00105         actuator_is_running_.remove(&async_exec.action_is_running);
00106     }
00107 }
```

```

00113 auto is_running() {
00114     std::lock_guard<std::mutex> lock(actuator_mutex_);
00115     actuator_is_running_();
00116
00117     auto result = false;
00118     for (const auto& ret : actuator_is_running_.results) {
00119         result = result || ret;
00120     }
00121     return result;
00122 }
00123
00129 static execution_poll& get() {
00130     static execution_poll instance;
00131     return instance;
00132 }
00133
00134 private:
00135 execution_poll() = default;
00136 ~execution_poll() = default;
00137 actuator<std::function<bool(void)>> actuator_is_running_;
00138 mutable std::mutex actuator_mutex_;
00139 };
00140
00154 template <typename actionT>
00155 class execution {
00156     // An execution may attach one of a different specialisation, which is a different class;
00157     // attach() and detach() reach into its attachment_lifetime_.
00158     template <typename otherActionT>
00159     friend class execution;
00160
00161 public:
00171     template <typename... Args>
00172     static std::shared_ptr<execution> create_instance(Args&&... args) {
00173         return std::make_shared<execution>(std::forward<Args>(args)...);
00174     }
00175
00181     execution() : execution(std::string(default_name)) {}
00182
00188     explicit execution(std::string exec_name) : name(std::move(exec_name)) {
00189         action_execute = untangle::bind(this, &execution<actionT>::execute_actions);
00190         action_stop = untangle::bind(this, &execution<actionT>::stop);
00191         action_is_running = untangle::bind(this, &execution<actionT>::is_running);
00192     }
00193
00208     ~execution() {
00209         // Out of the attacher first, before the worker is even asked to stop: from here on nothing
00210         // may reach this object, and the actions about to be destroyed are the ones it holds.
00211         if (!attachment_lifetime_ -> attacher.expired()) {
00212             attacher_execute_ -> remove(&action_execute);
00213             attacher_stop_ -> remove(&action_stop);
00214         }
00215
00216         {
00217             std::lock_guard<std::mutex> lock(action_mutex_);
00218             started_ = false;
00219             stopped_ = true;
00220         }
00221
00222         action_cv_.notify_all();
00223
00224         while (running_.load()) { // time of check
00225             std::this_thread::sleep_for(std::chrono::milliseconds(1));
00226         }
00227
00228         // time of use: this object is freed once the destructor returns, so the check above is only
00229         // safe because running_ is the last thing the worker touches
00230         execution_poll::get().remove(*this);
00231     }
00232
00250     template <typename classT, typename T>
00251     static void bind_action_and_method(actionT& action, const std::shared_ptr<classT>& obj,
00252                                       T classT::* method,
00253                                       const std::shared_ptr<execution>& async_exec) {
00254         auto async_action = std::make_shared<const actionT>(untangle::bind(obj, method));
00255
00256         action = [wp = std::weak_ptr<execution>(async_exec),
00257                 async_action](auto&&... args) -> actionT::result_type {
00258             // lock() also keeps the execution alive for the duration of the call
00259             const auto exec = wp.lock();
00260             if (!exec) {
00261                 throw invalid_action("bind: invalid execution");
00262             }
00263
00264             exec -> add_queued_action([async_action, ... args = std::forward<decltype(args)>(args)] {
00265                 return (*async_action)(args...);
00266             });
00267

```

```

00268     return typename actionT::result_type();
00269 };
00270 }
00271
00283 template <typename T>
00284 static void bind_action_and_function(actionT& action, T Fn,
00285                                     const std::shared_ptr<execution>& async_exec) {
00286     auto async_action = std::make_shared<const actionT>(std::move(Fn));
00287
00288     action = [wp = std::weak_ptr<execution>(async_exec),
00289             async_action](auto&&... args) -> actionT::result_type {
00290         const auto exec = wp.lock();
00291         if (!exec) {
00292             throw invalid_action("bind: invalid execution");
00293         }
00294
00295         exec->add_queued_action([async_action, ... args = std::forward<decltype(args)>(args)] {
00296             return (*async_action)(args...);
00297         });
00298
00299         return typename actionT::result_type();
00300     };
00301 }
00302
00314 bool is_running() const { return running_.load(); }
00315
00331 bool is_busy() const {
00332     std::lock_guard<std::mutex> lock(action_mutex_);
00333     return !action_list_.empty() || executing_action_.load();
00334 }
00335
00349 void run() {
00350     wait_thread_to_finish();
00351     collecting_results_ = true;
00352     thread_ = std::thread(&execution::execute, this);
00353     thread_.detach();
00354 }
00355
00366 void start() {
00367     wait_thread_to_finish();
00368     collecting_results_ = false;
00369     {
00370         std::lock_guard<std::mutex> lock(action_mutex_);
00371         started_ = true;
00372         stopped_ = false;
00373     }
00374
00375     thread_ = std::thread(&execution::loop, this);
00376     thread_.detach();
00377 }
00378
00385 void stop() {
00386     {
00387         std::lock_guard<std::mutex> lock(action_mutex_);
00388         started_ = false;
00389         stopped_ = true;
00390     }
00391
00392     // The worker drains what is already queued before it leaves loop(), so stop() does not have to
00393     // spin on the list; waking it is enough.
00394     action_cv_.notify_all();
00395
00396     if (actuator_stop_.is_connected()) {
00397         actuator_stop_();
00398     }
00399 }
00400
00422 template <typename... Args>
00423 bool add_action(actionT action, Args&&... args) {
00424     return add_queued_action(std::bind(std::move(action), std::forward<Args>(args)...));
00425 }
00426
00446 template <typename otherT>
00447 void attach(const std::shared_ptr<otherT>& other) {
00448     if (static_cast<const void*>(other.get()) == static_cast<const void*>(this)) {
00449         throw invalid_attachment("attach: an execution cannot be attached to itself");
00450     }
00451
00452     if (!other->attachment_lifetime_->attacher.expired()) {
00453         throw invalid_attachment("attach: that execution is attached already");
00454     }
00455
00456     // At most one attacher each makes the graph a forest, so a cycle closes exactly when other is
00457     // somewhere up this chain. Walking it is the whole check.
00458     for (auto link = attachment_lifetime_->attacher.lock(); link; link = link->attacher.lock()) {
00459         if (link == other->attachment_lifetime_) {
00460             throw invalid_attachment("attach: the attachment would close a cycle");

```

```

00461     }
00462 }
00463
00464 if (!actuator_execute_.is_connected()) {
00465     actuator_execute_ = untangle::connect(other->action_execute);
00466 } else {
00467     actuator_execute_.add(&other->action_execute);
00468 }
00469
00470 if (!actuator_stop_.is_connected()) {
00471     actuator_stop_ = untangle::connect(other->action_stop);
00472 } else {
00473     actuator_stop_.add(&other->action_stop);
00474 }
00475
00476 // What ~execution() needs to take itself back out of these actuators: the link proves this
00477 // attacher is still there, and the two pointers are only ever followed while it does.
00478 other->attachment_lifetime_->attacher = attachment_lifetime_;
00479 other->attacher_execute_ = &actuator_execute_;
00480 other->attacher_stop_ = &actuator_stop_;
00481 }
00482
00494 template <typename otherT>
00495 bool detach(otherT& other) {
00496     // The actuator's own action list is the record of what is attached; nothing else has to keep
00497     // one. An action is stored as a pointer, so the attachment is found by identity.
00498     const auto& actions = actuator_execute_.actions;
00499     if (std::find(actions.begin(), actions.end(), &other.action_execute) == actions.end()) {
00500         return false;
00501     }
00502
00503     actuator_execute_.remove(&other.action_execute);
00504     actuator_stop_.remove(&other.action_stop);
00505
00506     other.attachment_lifetime_->attacher.reset();
00507     other.attacher_execute_ = nullptr;
00508     other.attacher_stop_ = nullptr;
00509
00510     return true;
00511 }
00512
00524 auto results() const {
00525     std::lock_guard<std::mutex> lock(results_mutex_);
00526     return results_;
00527 }
00528
00529 static constexpr auto default_name = "default_name";
00530
00536 std::function<void(void)> action_execute;
00537
00545 std::function<void(void)> action_stop;
00546
00555 std::function<bool(void)> action_is_running;
00556
00562 std::function<void(void)> on_finished;
00563 std::string name = default_name;
00564
00565 private:
00566     using queued_action_t = std::function<typename actionT::result_type(void)>;
00567
00577 bool add_queued_action(queued_action_t action) {
00578     {
00579         std::lock_guard<std::mutex> lock(action_mutex_);
00580
00581         // Once stopped, the worker is on its way out and would never reach this action; dropping it
00582         // here is what keeps it from sitting in the list looking as though it were pending.
00583         if (stopped_) {
00584             std::println(stderr, "warning: execution '{}' is stopped, action not added", name);
00585             return false;
00586         }
00587
00588         action_list_.push_back(std::move(action));
00589     }
00590
00591     action_cv_.notify_one();
00592     return true;
00593 }
00594
00601 void execute_action(queued_action_t& action) {
00602     if constexpr (std::is_void_v<typename actionT::result_type>) {
00603         action();
00604     } else {
00605         auto value = action();
00606
00607         if (collecting_results_) {
00608             std::lock_guard<std::mutex> lock(results_mutex_);
00609             results_.push_back(std::move(value));

```

```

00610     }
00611   }
00612 }
00613
00623 void execute_actions() {
00624     actions_run_ = 0;
00625
00626     // The action is taken off the list under the lock and invoked with the lock released: an action
00627     // is caller code that may run for a while, and may itself call add_action().
00628     for (;;) {
00629         queued_action_t action;
00630
00631         {
00632             std::lock_guard<std::mutex> lock(action_mutex_);
00633             if (action_list_.empty()) {
00634                 break;
00635             }
00636
00637             action = std::move(action_list_.front());
00638             action_list_.pop_front();
00639
00640             // Under the lock that emptied the list, so is_busy() sees the pop and this together or
00641             // neither.
00642             executing_action_ = true;
00643         }
00644
00645         // Nothing may leave this loop: an exception escaping a detached thread function calls
00646         // std::terminate.
00647         try {
00648             execute_action(action);
00649         } catch (const invalid_action& ia) {
00650             std::println(stderr, "warning: execution '{}'" dropped an invalid action: {}", name,
00651                 ia.what());
00652         } catch (const std::exception& e) {
00653             std::println(stderr, "warning: execution '{}'" dropped an action that threw: {}", name,
00654                 e.what());
00655         } catch (...) {
00656             std::println(stderr, "warning: execution '{}'" dropped an action that threw an unknown type",
00657                 name);
00658         }
00659
00660         executing_action_ = false;
00661
00662         // Counted whether it ran cleanly, reported a dead binding, or threw: it came off the queue,
00663         // and the queue is what the notification is about.
00664         ++actions_run_;
00665
00666         // The notification is raised without the lock: on_finished may call add_action(), which
00667         // takes it.
00668         bool drained = false;
00669         {
00670             std::lock_guard<std::mutex> lock(action_mutex_);
00671             drained = action_list_.empty();
00672         }
00673
00674         if (drained) {
00675             notify_finished();
00676         }
00677     }
00678
00679     // Last, and part of the pass: an attachment is only ever reached through action_execute, which
00680     // is this function, so whoever drives this execution drives the ones attached to it.
00681     if (actuator_execute_.is_connected()) {
00682         actuator_execute_();
00683     }
00684 }
00685
00690 void notify_finished() {
00691     if (actions_run_ == 0 || !on_finished) {
00692         return;
00693     }
00694
00695     {
00696         // An action may queue another, so a pass that drained can leave more behind it. That is the
00697         // next batch, not the end of this one.
00698         std::lock_guard<std::mutex> lock(action_mutex_);
00699         if (!action_list_.empty()) {
00700             return;
00701         }
00702     }
00703
00704     on_finished();
00705 }
00706
00714 void wait_thread_to_finish() {
00715     auto resident = false;
00716     while (!running_.compare_exchange_weak(resident, true)) {

```

```

00717     resident = false;
00718     std::this_thread::sleep_for(std::chrono::milliseconds(1));
00719 }
00720 }
00721
00722 void execute() {
00723     // The results belong to this run, not to this run and the last one.
00724     {
00725         std::lock_guard<std::mutex> lock(results_mutex_);
00726         results_.clear();
00727     }
00728
00729     execute_actions();
00730
00731     std::println("execution '{}'' finishing thread", name);
00732
00733     // Must stay last: ~execution() may free this object the moment it reads false.
00734     running_ = false;
00735 }
00736
00737 void loop() {
00738     for (;;) {
00739         {
00740             std::unique_lock<std::mutex> lock(action_mutex_);
00741
00742             // Bounded: an attached execution has its own list and cannot notify this condition
00743             // variable.
00744             action_cv_.wait_for(lock, std::chrono::milliseconds(10),
00745                 [this] { return !action_list_.empty() || !started_; });
00746
00747             if (!started_ && action_list_.empty()) {
00748                 break;
00749             }
00750         }
00751         execute_actions();
00752     }
00753
00754     // A last pass for the attachments; this execution's own list is already empty here.
00755     execute_actions();
00756
00757     std::println("execution '{}'' thread finished", name);
00758
00759     // Must stay last: ~execution() may free this object the moment it reads false.
00760     running_ = false;
00761 }
00762
00763 std::thread thread_;
00764
00765 std::atomic_bool started_ = {false};
00766 std::atomic_bool running_ = {false};
00767
00768 // Set by stop(), and the difference between "not started yet" and "finished for good": actions
00769 // may be queued before run()/start(), but not after stop().
00770 std::atomic_bool stopped_ = {false};
00771
00772 // action_list_, started_ and stopped_ are written by every thread that calls add_action() or
00773 // stop() and read by the worker; nothing touches them outside this mutex.
00774 mutable std::mutex action_mutex_;
00775 std::condition_variable action_cv_;
00776
00777 std::list<std::function<typename actionT::result_type(void)>> action_list_;
00778
00779 using void_actuator = actuator<std::function<void(void)>>;
00780
00781 void_actuator actuator_execute_;
00782 void_actuator actuator_stop_;
00783
00784 std::shared_ptr<attachment> attachment_lifetime_ = std::make_shared<attachment>();
00785
00786 void_actuator* attacher_execute_ = nullptr;
00787 void_actuator* attacher_stop_ = nullptr;
00788
00789 // std::vector cannot hold void type; use an arbitrary type e.g. int
00790 using resultT = std::conditional<std::is_void<typename actionT::result_type>::value, int,
00791     typename actionT::result_type>;
00792
00793 std::vector<typename resultT::type> results_;
00794
00795 mutable std::mutex results_mutex_;
00796
00797 std::atomic_bool collecting_results_ = {false};
00798
00799 std::atomic_bool executing_action_ = {false};
00800
00801 std::atomic_size_t actions_run_ = {0};
00802 };
00803 // namespace async

```

```
00859 } // namespace untangle
```


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